CSCI 611 Assignment 2

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1. Model Architecture and Training Setup

This project trains a custom convolutional neural network using PyTorch to classify images from the CIFAR 10 dataset. CIFAR 10 contains 60000 color images of size 32 by 32 across 10 classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The goal is to build a non pretrained CNN model, train it end to end, evaluate classification performance on the test set, and interpret learned features through visualization.

The implemented CNN uses three convolutional layers followed by max pooling and ReLU nonlinearity. Dropout is included to reduce overfitting. The convolutional feature extractor is followed by a fully connected classification head that outputs logits for the 10 classes.

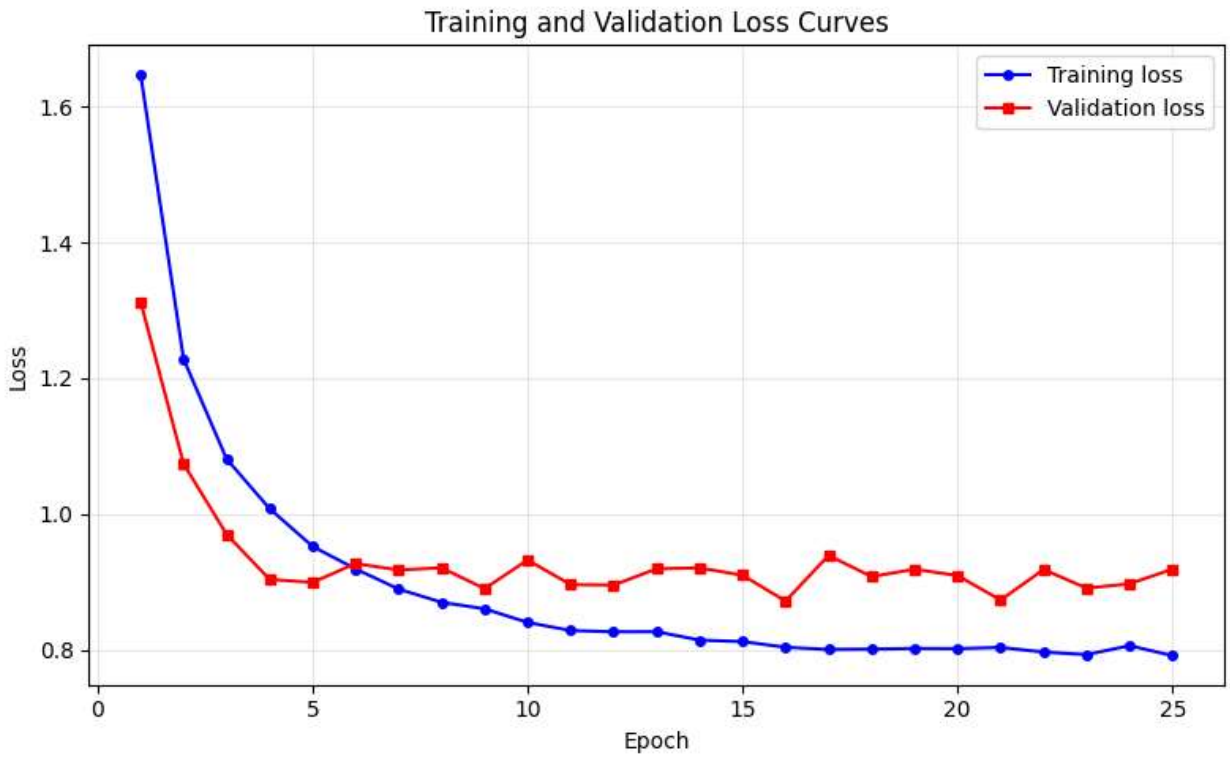
Training uses cross entropy loss for multi class classification. The optimizer is stochastic gradient descent with momentum and weight decay. Training is performed for 25 epochs using a batch size of 20. The best model checkpoint is selected based on the lowest validation loss and saved for final testing.

Training configuration used in this experiment is as follows.
Loss function: CrossEntropyLoss
Optimizer: SGD
Learning rate: 0.01
Momentum: 0.9
Weight decay: 1e 4
Batch size: 20
Epochs: 25
Dropout probability: 0.25

2. Training Results

Training and validation loss were monitored over 25 epochs. The training loss decreased steadily from approximately 1.65 in the first epoch to roughly 0.80 near the end of training. Validation loss dropped quickly during early epochs and then stabilized around 0.90 to 0.93 with small oscillations. This pattern suggests the model learned useful features rapidly and then reached a regime where additional training primarily improved training loss more than validation loss, indicating mild overfitting. The dropout and weight decay helped limit

overfitting, but the gap between training and validation loss still becomes noticeable after the first several epochs.



The final saved model was evaluated on the CIFAR 10 test set. The overall test loss is 0.903694 and the overall test accuracy is 69 percent, corresponding to 6950 correct predictions out of 10000 test images. This meets the expected performance range for a simple custom CNN on CIFAR 10.

Per class test accuracy is shown below.
Airplane 72 percent 722 out of 1000
Automobile 84 percent 848 out of 1000
Bird 62 percent 625 out of 1000
Cat 59 percent 590 out of 1000
Deer 70 percent 707 out of 1000
Dog 48 percent 484 out of 1000
Frog 74 percent 748 out of 1000
Horse 68 percent 687 out of 1000
Ship 79 percent 796 out of 1000
Truck 74 percent 743 out of 1000

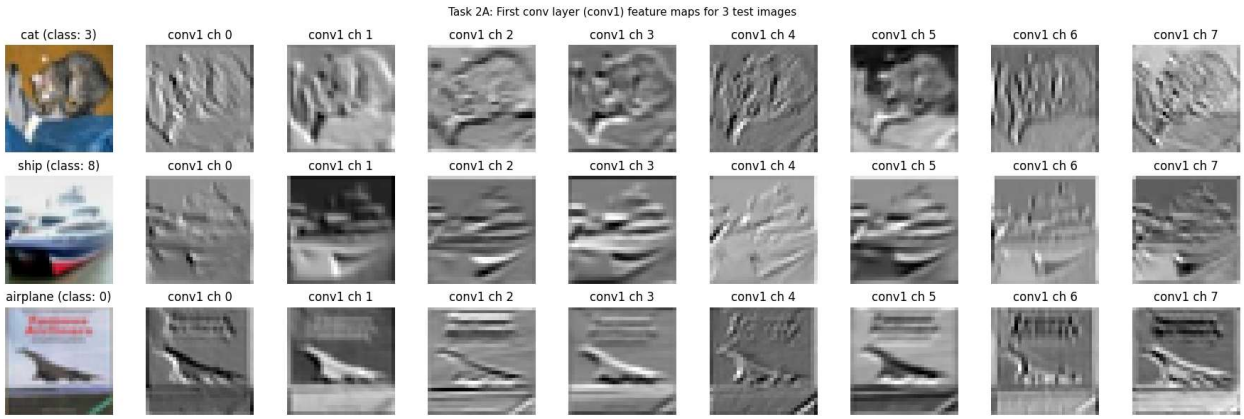
The strongest class performance occurs for automobile and ship, while the weakest occurs for dog. A likely explanation is that dogs in CIFAR 10 often appear at different scales and poses and can resemble cats or other mammals under low resolution, making them harder to separate with a compact CNN. The sample prediction visualization also shows several plausible confusions such as cat predicted as dog and deer predicted as frog.

3. Feature Map Visualization Early Layer

To interpret what the network learns in the early stages, feature maps from the first convolutional layer were visualized for three test images from different classes. The first convolutional layer tends to learn low level detectors such as oriented edges, color contrast boundaries, and simple texture patterns. This is expected because early layers process raw pixels and respond strongly to local gradients and color changes.

In the feature map visualization, several channels respond strongly to prominent edges and boundaries. For the cat example, multiple channels highlight the outline of the head and ear edges as well as texture patterns in the fur. For the ship example, some channels emphasize horizontal edges and structured patterns consistent with the ship body and

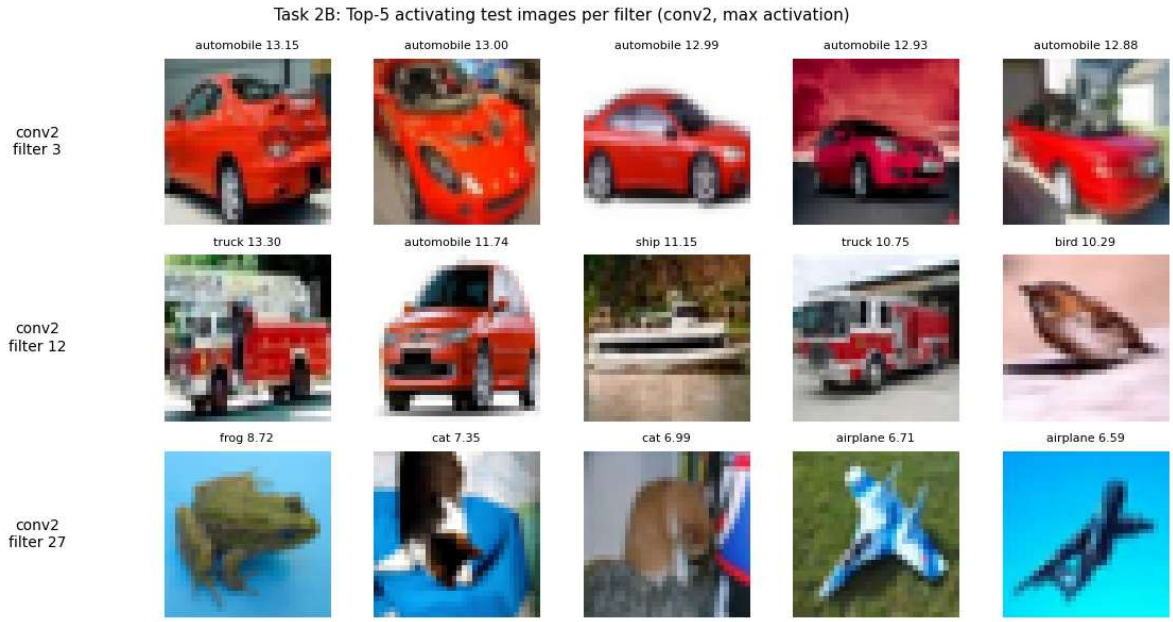
background separation. For the airplane example, channels respond to the airplane silhouette and high contrast regions between the plane and the sky or background text. Differences across channels show that each filter specializes in a distinct pattern, with some focusing on sharp boundaries and others capturing broader intensity transitions.



4. Maximally Activating Images

A second visualization approach finds the test images that most strongly activate specific filters in a chosen convolutional layer. This provides insight into what visual patterns each filter is tuned to detect. In this experiment, three filters from the second convolutional layer were analyzed. Activation was defined using the maximum value over the two dimensional feature map produced by the filter for each image. Each test image was scored by this activation measure, images were ranked, and the top five images were selected per filter.

The top activating images show meaningful structure. For the selected filter 3 in conv2, the top images are predominantly automobiles with strong red color regions and curved body shapes, suggesting the filter may respond to combinations of color and shape patterns common in vehicles. For filter 12, the top images include trucks and automobiles and one ship and bird, indicating the filter may be capturing high contrast block like structures and edges rather than a single class specific feature. For filter 27, the top images include frog, cat, airplane, and other classes, suggesting the filter responds to a general texture or contour pattern that appears across multiple classes rather than a specific object category.



5. Brief Discussion and Reflection

This assignment demonstrates that a compact custom CNN with three convolutional layers can achieve solid performance on CIFAR 10. The model achieves 69 percent overall test accuracy with strong results for automobile and ship and weaker results for dog and cat, likely due to intra class variation and similarity between small animal classes at low resolution. The training curves show rapid learning in early epochs and a later regime where validation loss stabilizes while training loss continues to decrease, suggesting mild overfitting.

The visualization tasks provide interpretability. Feature maps from the first layer confirm that early filters detect edges, contours, and simple textures. The maximally activating image analysis shows that mid layer filters respond to more complex combinations of local patterns, sometimes aligning with object classes such as vehicles, and sometimes capturing more general textures or structural cues shared across categories.

Given more time, additional improvements could include stronger data augmentation, deeper architectures with batch normalization, and learning rate scheduling to improve generalization and reduce the gap between training and validation loss. Another useful extension would be computing a confusion matrix to quantify which classes are most frequently confused and to relate those confusions to the learned filter responses.